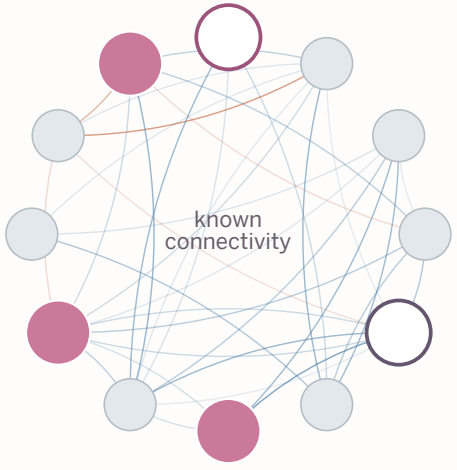
Prior-averaged risk often leaves recording decisions unchanged

Choosing 3 neurons to record in order to predict a stimulus response; simulated 12-neuron networks with known wiring



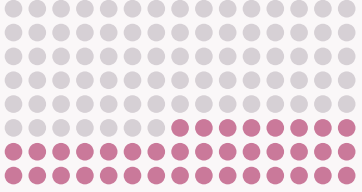
- recorded node (M = 3)
- stimulated node
- response readout

$$q = \int [r_{\text{stim}}(t) - r_{\text{control}}(t)]\, dt$$

1 Recording agreement

Nominal vs prior-averaged risk
40 independent networks

68% 82 of 120 single-target designs are identical



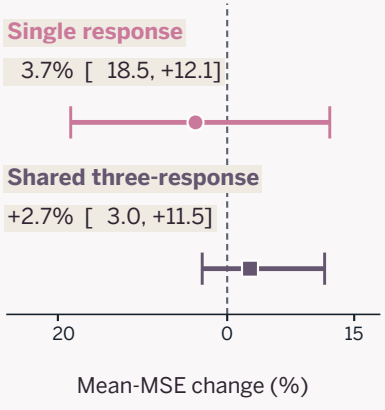
65% 26 of 40 shared-target designs are identical



same recording set

2 Response error

Common MAP fit and budget
40 independent networks

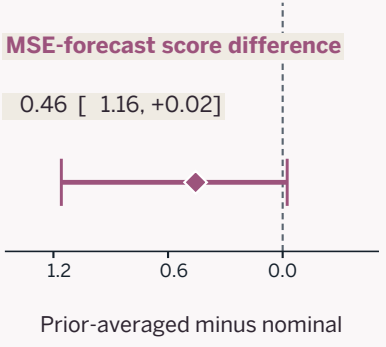


97.5% paired network-cluster CI

3 Risk forecasts

Fixed data and point prediction
40 new independent networks

19 of 40 networks score lower; 21 score higher



95% paired network-cluster CI

PRIMARY AND EXPLORATORY FINDINGS

Identical recordings yield identical predictions under the common fit. Primary error and forecast intervals include zero.

In the 100-unit extension, 4 of 8 intervals exclude zero, all favouring prior averaging (8 networks; exploratory).

Extension: unadjusted 95% intervals; one point estimate reverses sign. All quantities are simulated.